AI-Driven Content Strategy Engine

Reverse-engineering what wins in a niche, then generating content designed to perform

DSC680 — Applied Data Science | Project 2 | Milestone 1: Project Proposal

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1. Topic

Topic: Build a data-driven content strategy and generation pipeline that analyzes 10,000–15,000 niche-specific social posts and articles, identifies the topics, formats, hooks, and posting patterns that statistically predict engagement, and uses a large language model to generate brand-aligned content drafts predicted to perform within that niche.

Content marketing is a guessing game for most small and mid-sized businesses. Generative AI tools draft posts quickly, but they do not answer the harder question — what should we say in the first place? This project closes that gap by combining statistical analysis of public engagement signals with LLM-based generation, producing a defensible content playbook and 20 draft pieces tied to evidence rather than intuition.

2. Business Problem and Research Questions

2.1 Why this matters

Industry estimates put the global content marketing market at roughly $107 billion in 2026, with longer-horizon forecasts reaching $1.95 trillion by 2032 (Content Marketing Institute, 2026). Content is now the single largest line item in most marketing organizations, at about 26% of total marketing spend, yet measuring and predicting content ROI remains the most-cited challenge for marketing leaders. AI generation tools such as Jasper and Copy.ai have lowered the cost of producing content but have not lowered the cost of producing the wrong content. Marketing teams ship more posts than ever and cannot reliably explain why some succeed and most do not.

The proposed system targets this asymmetry. It treats publicly available engagement data — Reddit upvotes and comments, blog social-share counts, and Google Trends search demand — as a revealed-preference signal for what audiences in a given niche actually engage with. Patterns extracted from that signal then condition an LLM generator so that the content produced for a client is grounded in observed engagement, not stylistic preference.

Primary stakeholders include in-house marketing teams at direct-to-consumer brands and B2B SaaS companies, content agencies, and solo marketing consultants who currently rely on intuition or recycled best-practice posts. The deliverable for these stakeholders is a content playbook — ranked themes, optimal formats, posting cadence — plus a batch of generated drafts tagged with a predicted engagement tier and a short rationale.

2.2 Research questions

1. RQ1: Which content features — topic category, format (how-to, listicle, opinion, case study), hook type (question, statistic, story, contrarian), length, and posting time — most strongly predict engagement in a target niche, and what are the magnitudes (effect sizes) of these relationships after controlling for poster or publication size?
2. RQ2: Do mean engagement levels differ significantly across topic categories and content formats within a niche (one-way ANOVA with post-hoc comparisons and η^2 effect size), and how large is the practical difference?
3. RQ3: Can Google Trends search demand time series serve as a leading indicator of high-engagement topics on social platforms, and is there a measurable lead time between search-interest inflection and on-platform engagement uplift?
4. RQ4: Does LLM-generated content conditioned on the empirically derived patterns score higher on a held-out engagement-prediction model than baseline LLM output written without those constraints, and what is the bootstrap confidence interval around that difference?

RQ1–RQ2 produce the strategy layer (what works). RQ3 adds a forward-looking dimension (what is about to work). RQ4 closes the loop by validating that the generation pipeline operationalizes the strategy.

3. Datasets

All data sources are public, free to access for academic coursework, and contain no protected personal information when used at the aggregate level described below. Specific platform terms of service and rate limits are respected throughout collection.

3.1 Reddit (primary source)

- **Access:** Reddit Data API via the PRAW Python wrapper, free authenticated tier.
- **Scope:** 10–15 niche-specific subreddits (the candidate niche for this project is direct-to-consumer fitness and recovery; a backup niche is B2B SaaS marketing). Target sample of 10,000–15,000 posts, sampled across at least the most recent 12 months to capture seasonality.
- **What each row represents:** one Reddit submission, with fields for upvotes, score, number of comments, awards, post type (text, link, image), submission timestamp, subreddit, title text, and (where present) body text.
- **Why appropriate for coursework:** Public posts with no expectation of private analysis at the user level. Analysis is aggregate, focused on content attributes (topic, format, hook, length, time) rather than identifying individual users. Author handles are dropped

after a sub-level control is computed; no PII or sensitive demographic inference is performed.

3.2 Industry blog and article corpus (secondary source)

- **Access:** Targeted scraping of 20–30 industry blogs using BeautifulSoup, with robots.txt honored and a polite rate limit. Target sample of 500–1,000 articles.
- **What each row represents:** one article, with fields for title, publish date, word count, on-page meta description, topic tags (where present), and publicly displayed social share counts where the site exposes them.
- **Why appropriate:** Provides a long-form counterpart to Reddit so the analysis is not platform-bound. Only metadata and short summaries are stored; full article bodies are not redistributed.

3.3 Google Trends (forward-looking signal)

- **Access:** trendspyg (an actively maintained successor to the now-archived pytrends library) against the public Google Trends interface, free.
- **What each row represents:** weekly search-interest index for a topic-keyword bundle (top 50 candidate topics derived from RQ1 features), plus related-queries and rising-queries data.
- **Why appropriate:** Adds an exogenous demand signal independent of the platforms whose engagement is being modeled, which lets RQ3 test a lead-lag relationship rather than only retrospective correlation.

All three sources are documented on their respective public pages and have been used in peer-reviewed marketing and information-science research, supporting their suitability for academic reuse.

4. Methods

The pipeline runs in five stages: collect, feature-engineer, model, generate, evaluate. Each stage maps to coursework already completed in the MSDS program (DSC500 statistical fundamentals, DSC540 data preparation, DSC630 predictive analytics, DSC670 generative AI, DSC640 visualization).

4.1 Data collection

- Reddit submissions collected with PRAW; pagination respected, request rate held below the documented Reddit API ceiling, and each call wrapped with exponential backoff on transient failures.
- Blog corpus collected with BeautifulSoup over a polite scraper that honors robots.txt and caches HTML locally so analysis can be reproduced without re-hitting the source sites.

- Google Trends pulled via trendspyg in weekly intervals over a 24-month window to support both seasonality detrending and lead-lag analysis. (pytrends, the previously dominant client, was archived in April 2025, so the project uses an actively maintained successor.)

4.2 Feature engineering

- Structural features: word count, title length, presence of digits or numerals, presence of question mark, day-of-week, hour-of-day in the subreddit's modal time zone.
- LLM-tagged features: topic category (closed taxonomy of 12 niche-relevant topics), format (how-to, listicle, opinion, case study, news, question, story), hook type (question, statistic, story, contrarian, direct claim), and overall tone.
- Control variables: subreddit subscriber count at collection time, post age in days at collection time, and publication domain authority proxy for blog rows (logged share count percentile within the publication).

4.3 Statistical and predictive modeling

- RQ1 — engagement prediction. Outcome variables are log(1 + upvotes) and log(1 + comments) to handle the heavy right tail. Models compared: ordinary least squares with robust standard errors, gradient-boosted regression (XGBoost), and a random forest. Feature importance via permutation importance and SHAP values (Lundberg & Lee, 2017) so direction and magnitude of each driver are interpretable, not just predictive.
- RQ2 — categorical comparison. One-way ANOVA across topic categories and across formats, with Tukey HSD post-hoc pairwise tests and η^2 effect size (Cohen, 1988) to keep the practical-versus-statistical-significance distinction explicit.
- RQ3 — lead-lag detection. Cross-correlation between weekly Google Trends interest and weekly engagement volume for the top 20 topics, with Granger causality tests as a supporting check. Reporting will state explicitly that Granger causality is a temporal-precedence test rather than evidence of mechanism.
- RQ4 — generation evaluation. Twenty AI-generated drafts per niche are scored by the engagement-prediction model trained in RQ1, then compared to a matched baseline of LLM drafts produced without the empirically derived constraints. Difference in mean predicted engagement is reported with a 1,000-resample bootstrap 95% confidence interval (Efron & Tibshirani, 1993).
- A/B variant generation. For each draft, two variants are produced (different hook, different angle) and a power analysis is included so a client running the live test can compute the sample size needed to detect a meaningful uplift at $\alpha = 0.05$ and 80% power.

4.4 Deliverables

- Streamlit dashboard — engagement heatmap (topic × format × day-of-week), top features bar chart with SHAP magnitudes, Google Trends lead-lag explorer, generated content calendar.
- Content playbook PDF — the strategic narrative that translates the model output into recommendations a non-technical client would act on.
- Reproducible code repository with README, requirements pin, and a single-command pipeline script.

5. Ethical Considerations

Findings in this project are associative and based on public engagement data. Engagement is not a measure of content quality or truthfulness, and the project frames its outputs accordingly.

- Privacy and aggregation. Although Reddit posts are public, users may not anticipate large-scale analysis of their behavior. All analysis is performed at the aggregate or content-attribute level; usernames are not published in any artifact, and no demographic inference about individual users is attempted.
- Platform terms and rate limits. Reddit Data API terms, target blogs' robots.txt directives, and Google Trends usage norms are followed. Caching is used to minimize repeated requests.
- Manipulation risk. A system that predicts engagement can be misused to optimize attention without regard for accuracy. The deliverable narrative will state that the playbook is a guide for legitimate marketing content and is not appropriate for political persuasion, health claims, or financial advice without separate domain review and disclosures.
- AI disclosure. LLM-generated drafts are flagged as such in the dashboard and accompanying client documentation, consistent with the updated U.S. Federal Trade Commission guidance on AI-augmented endorsements and synthetic content (FTC, 2026).
- Confounding and fairness. Engagement is shaped by poster size, platform algorithms, and time-of-day effects that are only partially observable. The write-up names these confounds rather than presenting predictions as causal. No sub-population fairness audit is conducted because no demographic features are modeled; that limitation is stated, not concealed.
- Intellectual property. Blog scraping captures titles, metadata, and short summaries only — not full article bodies — to avoid republishing copyrighted text. The generated content produced for clients is original output of the model, not a paraphrase of a single source article.

6. Anticipated Challenges

- Confounded engagement. Subreddit subscriber count, post age, and platform algorithm changes during the collection window will all push engagement in ways that are not driven by content features. Mitigation: explicit control variables, log-transformed outcomes, and effect-size reporting alongside p-values.
- Rate limits and scraper fragility. Reddit API and blog scrapers can fail mid-run. Mitigation: incremental writes to disk, resumable collection scripts, and a one-week buffer in the Week 1 timeline.
- LLM tagging noise. Topic and hook-type labels assigned by the LLM will not be perfect. Mitigation: a 200-row hand-labeled validation sample to estimate tag accuracy, plus a sensitivity analysis dropping low-confidence labels.
- Cross-platform engagement scales. Reddit upvotes and blog shares are not comparable directly. Mitigation: separate models per platform and a within-platform percentile rank used for any cross-platform visualization.
- Niche generalizability. Patterns found in one niche will not transfer wholesale to another. Mitigation: the proposal commits to one primary niche; transferability is discussed in the limitations section, not promised in results.
- Overstating prediction quality. Predicted engagement is uncertain, not deterministic. Mitigation: every predicted-tier label in the dashboard is accompanied by the model's prediction interval, and the playbook describes the system as decision support rather than autopilot.

7. References

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